Week 4

MICROECONOMIC THEORY
ECON 323 502/503

Spring 2026

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Review

1. Analytically, how to derive the demand curve of Good X:
   - Step 1. Fix P_Y and I. Set up a utility maximization problem.
   - Step 2. Solve the utility maximizing x as a function of P_X $\Rightarrow$ $Q_X(P_X)$

2. Analytically, how to derive the Engel curve of Good X:
   - Step 1. Fix P_X and P_Y. Set up a utility maximization problem.
   - Step 2. Solve the utility maximizing x as a function of I $\Rightarrow$ $Q_X(I)$

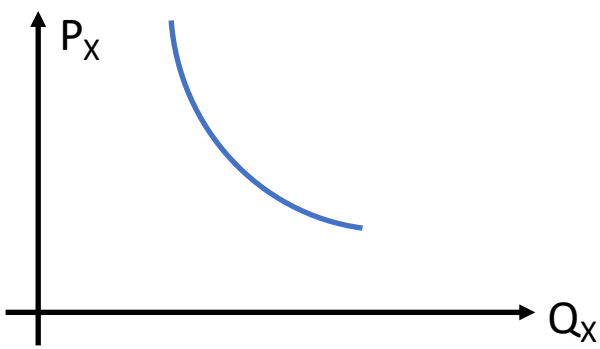
3. Read the relationship between P_Y and Q_X (or P_X and Q_Y) from the price-consumption curve. Two goods are
   - **substitutes**: if P_Y and Q_X change in the same direction
   - **complements**: if P_Y and Q_X change in the opposite direction
   - **independent**: if P_Y does not affect Q_X

4.2 Income and Substitution Effects

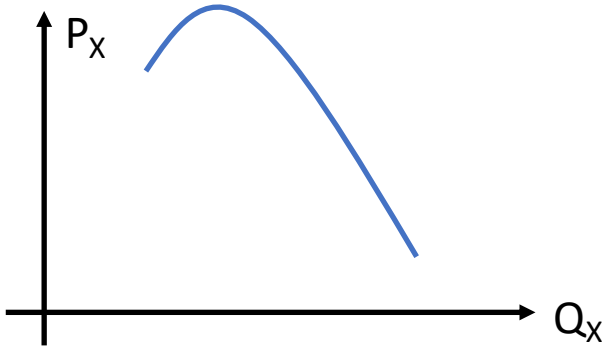
Take a closer look at the demand curve.

Usually, the **law of demand** holds: when the price of a good decreases (resp. increases), the quantity demanded of that good increases (resp. decreases).

The law of demand implies that the demand curve is downward sloping.



satisfies the law of demand



violates the law of demand

4.2 Income and Substitution Effects

- Does the law of demand always hold?
- Tentative answer: No.

- Is there an example where the law of demand is violated?
- Yes. Giffen good violates the law of demand.

- Why is the law of demand usually true? What can potentially make the law of demand go wrong?
- We will wait to analyze the substitution effect and income effect to answer this question.

4.2 Income and Substitution Effects

A fall in the price of a good has two effects:

- The consumer tends to buy more of the good that has become cheaper and less of the good that is relatively more expensive.
- The response to a change in the relative prices of goods is called the **substitution effect**.

- The consumer experiences an increase in real purchasing power, which affects her purchasing decision.
- The change in consumption resulting from the change in real purchasing power is called the **income effect**.

4.2 Income and Substitution Effects

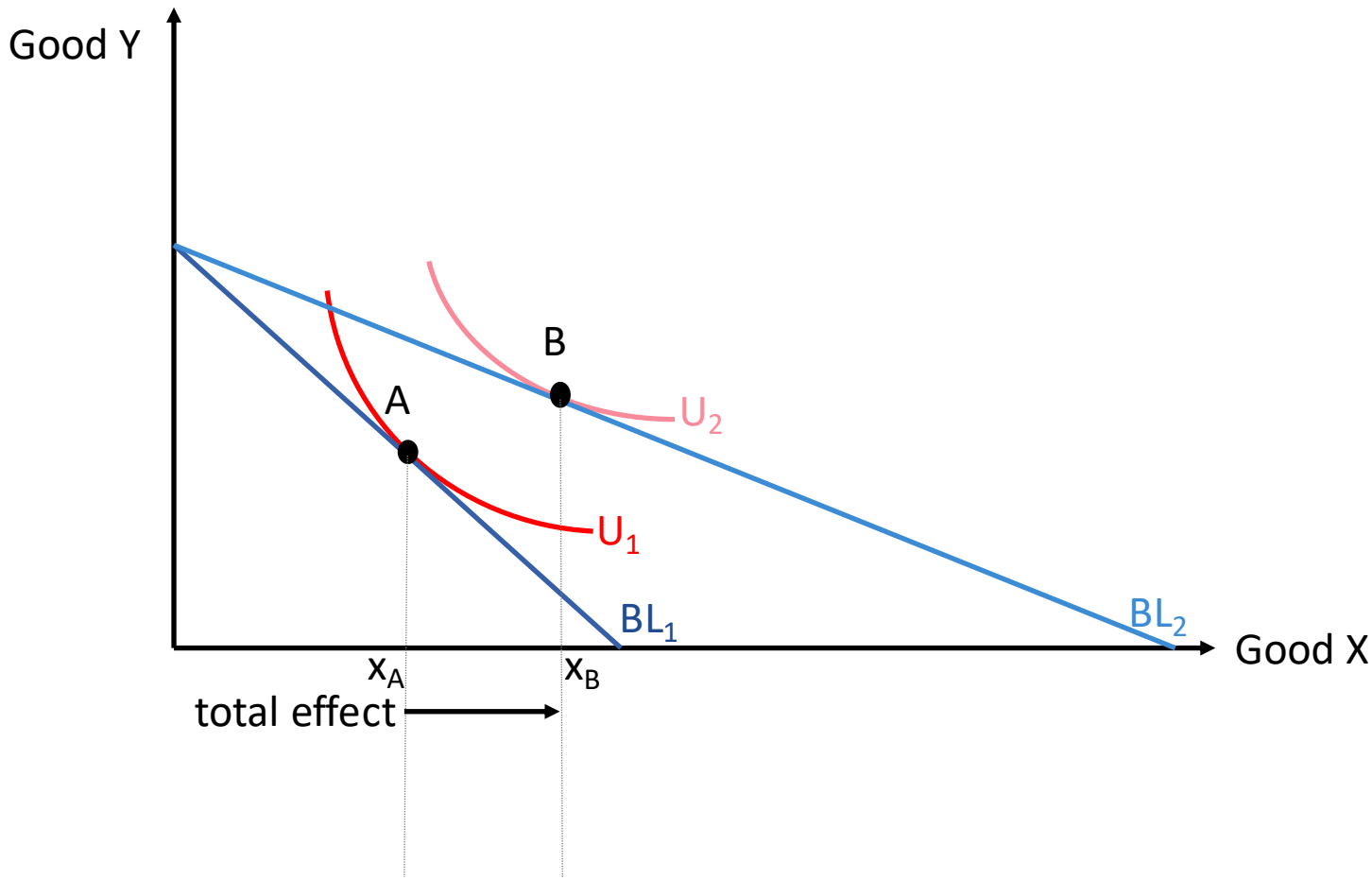
When there is a decrease in P_X, the two effects happen at the same time, but for analysis purpose, we will graphically disentangle the two effects.

We now consider three cases:

- Case 1: When Good X is a normal good.
- Case 2: When Good X is an inferior good with relatively small (in magnitude) income effect.
- Case 3: When Good X is an inferior good with relatively big (in magnitude) income effect.

4.2 Income and Substitution Effects (P_X decreases)

Case 1: Good X is a normal good.



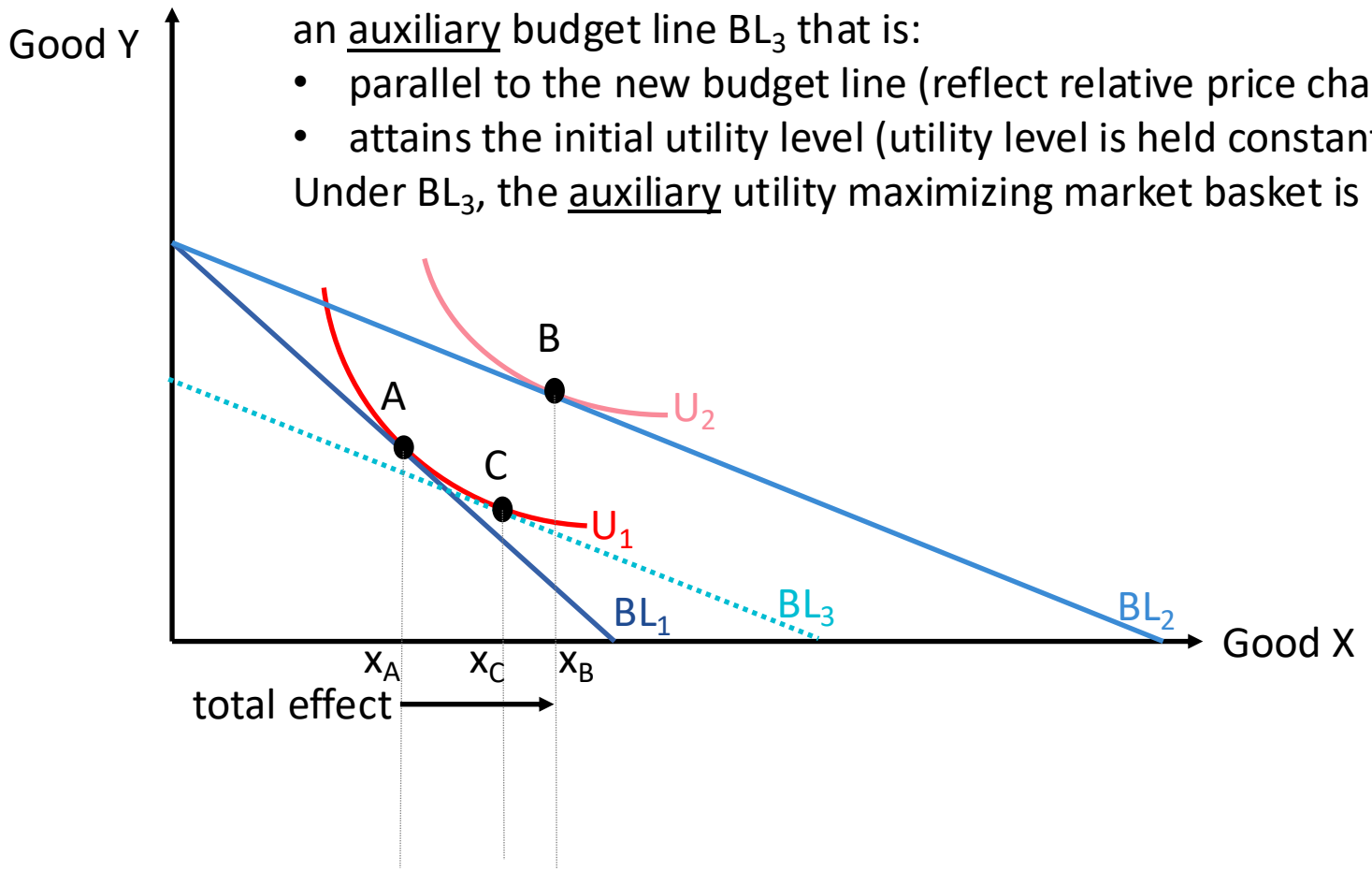
4.2 Income and Substitution Effects (P_X decreases)

Case 1: Good X is a normal good.

To disentangle the substitution effect and the income effect, we plot an <u>auxiliary</u> budget line BL_3 that is:

- parallel to the new budget line (reflect relative price change)
- attains the initial utility level (utility level is held constant)

Under BL_3, the <u>auxiliary</u> utility maximizing market basket is C.

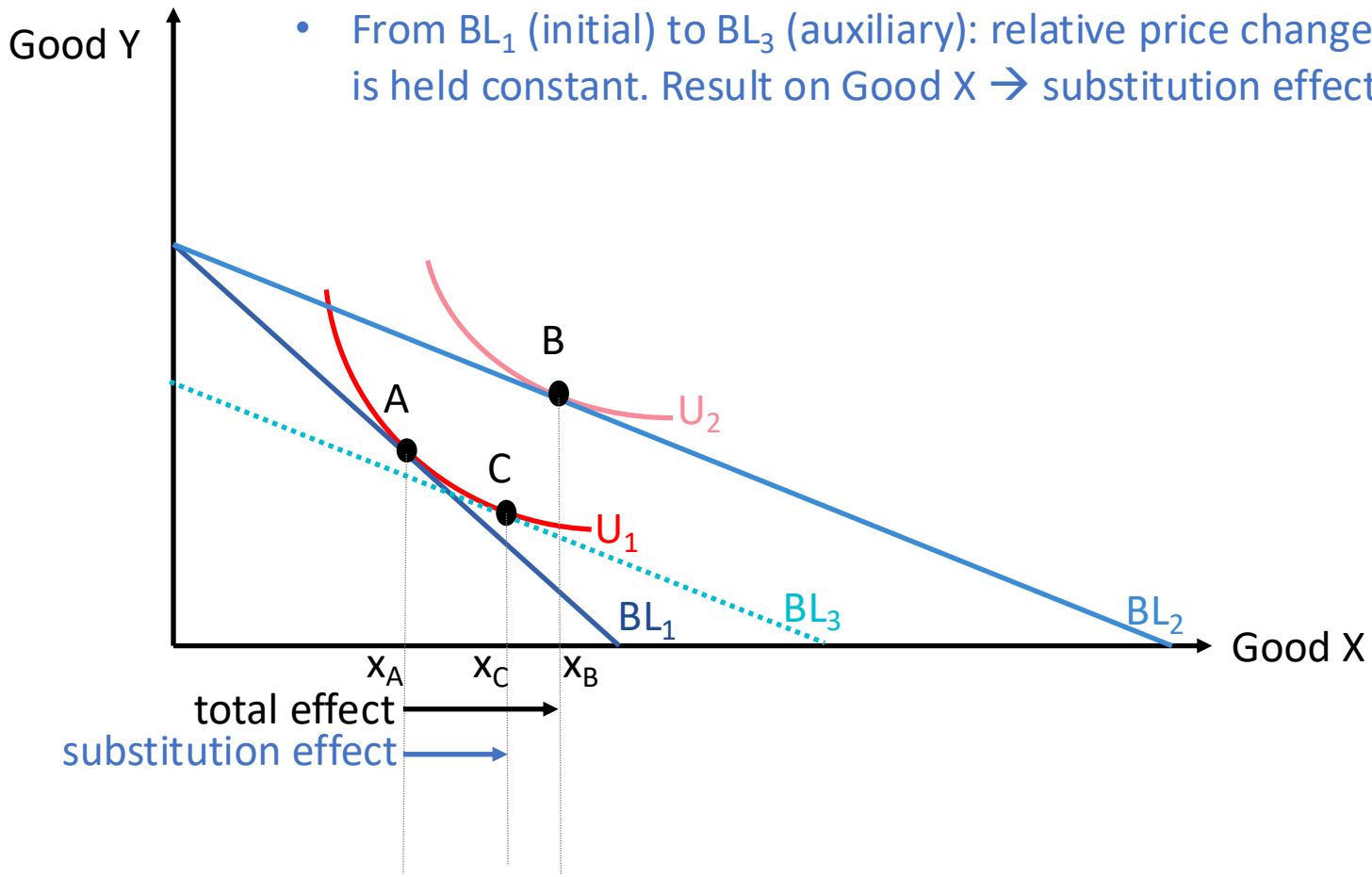


4.2 Income and Substitution Effects (P_X decreases)

Case 1: Good X is a normal good.

Breakdown the change from BL_1 (initial) to BL_2 (new) into two steps

- From BL_1 (initial) to BL_3 (auxiliary): relative price changes, utility level is held constant. Result on Good X → substitution effect.

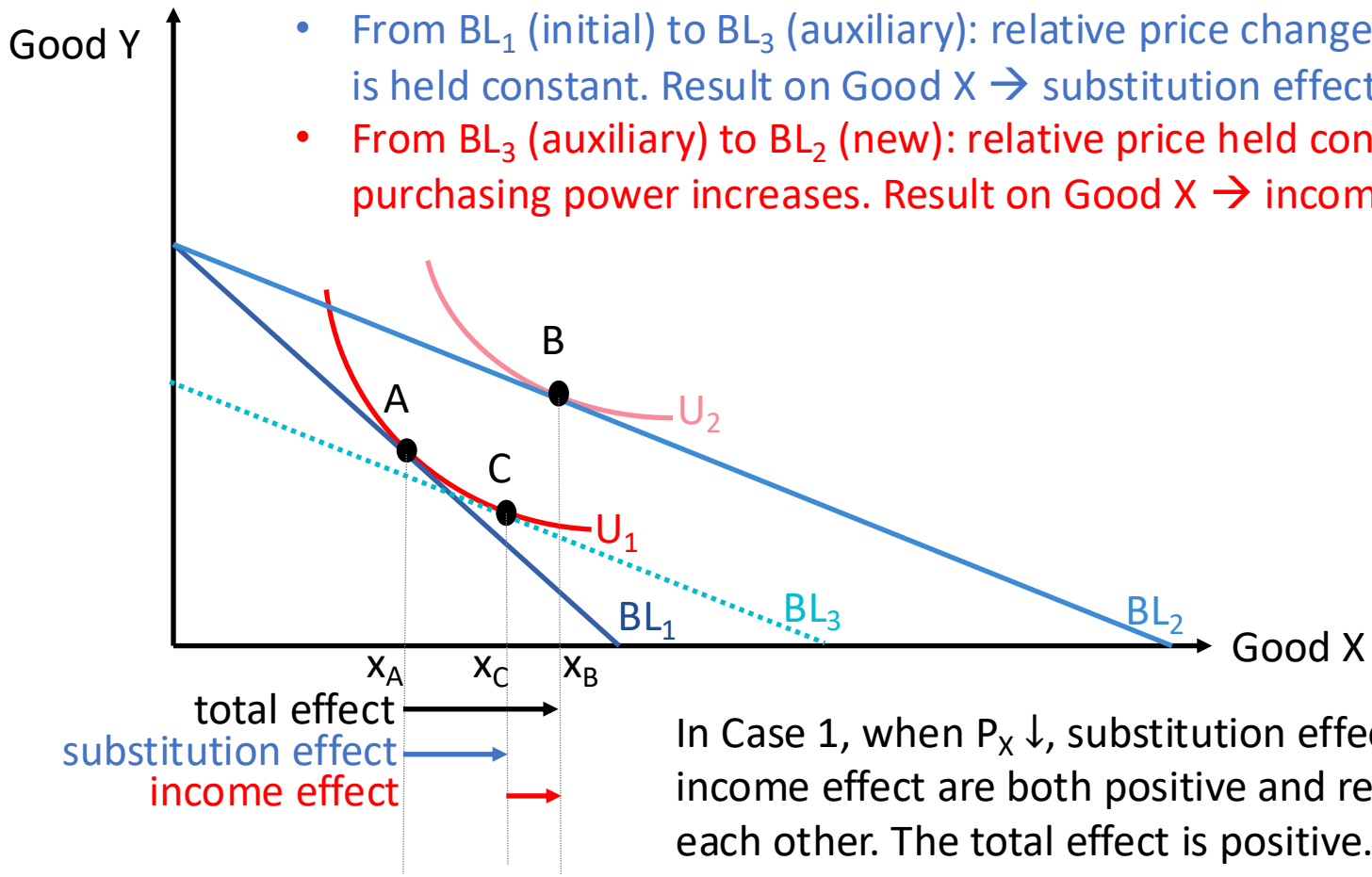


4.2 Income and Substitution Effects (P_X decreases)

Case 1: Good X is a normal good.

Breakdown the change from BL_1 (initial) to BL_2 (new) into two steps

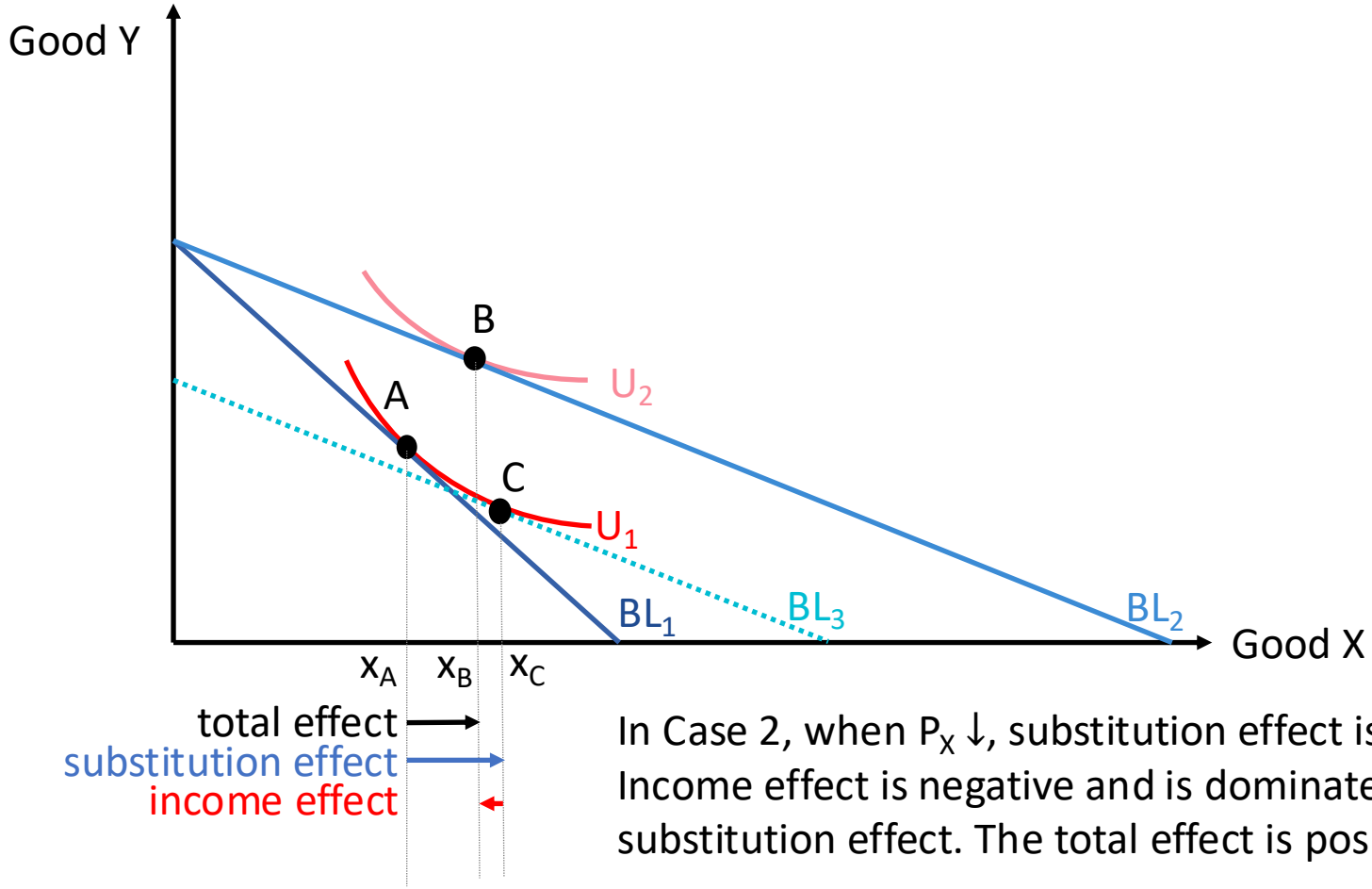
- From BL_1 (initial) to BL_3 (auxiliary): relative price changes, utility level is held constant. Result on Good X → substitution effect.
- From BL_3 (auxiliary) to BL_2 (new): relative price held constant, real purchasing power increases. Result on Good X → income effect.



In Case 1, when P_X ↓, substitution effect and income effect are both positive and reinforce each other. The total effect is positive.

4.2 Income and Substitution Effects (P_X decreases)

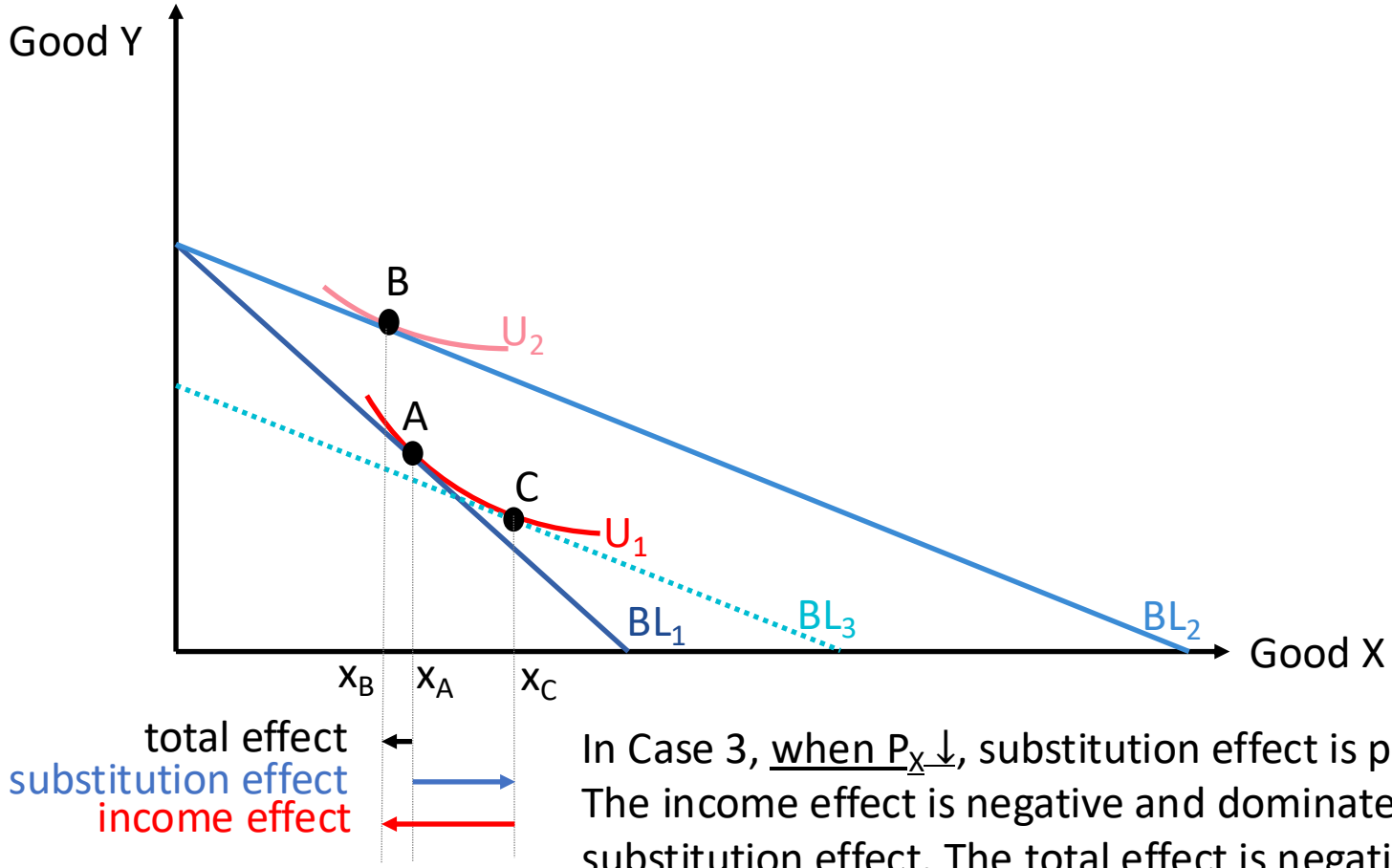
Case 2: Good X is an inferior good with relatively small (in magnitude) income effect.



In Case 2, when $P_X \downarrow$, substitution effect is positive. Income effect is negative and is dominated by the substitution effect. The total effect is positive.

4.2 Income and Substitution Effects (P_X decreases)

Case 3: Good X is an inferior good with relatively big (in magnitude) income effect.



In Case 3, <u>when $P_X \downarrow$</u>, substitution effect is positive. The income effect is negative and dominates the substitution effect. <u>The total effect is negative</u>.

4.2 Income and Substitution Effects

Remarks:

- In the third case, Good X is a **Giffen good**: it is a good for which the price of the good and the quantity demanded can change in the same direction, because the income effect moves in the opposite direction with the substitution effect and dominates the latter.

- Giffen good violates the law of demand, and can have an upward-sloping portion in the demand curve.

- **A Giffen good is a very inferior inferior good**.

Do Giffen goods exist in reality?

In the past, economists believed that Giffen goods exist in theory only. However, in mid 2000, two economists identified the existence of Giffen goods in a field experiment.

They find that for poor households in Hunan, China, rice is a Giffen good. For poor households in Gansu, China, wheat is a Giffen good.

Receive discount coupons for rice→ Buy less rice (more meat)

Reference:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2964162/

4.2 Income and Substitution Effects

- Does the law of demand always hold?
- Answer: No.

- Is there an example where the law of demand is violated?
- Answer: Yes. Giffen good violates the law of demand.

- Why is the law of demand usually true? What can potentially make the law of demand go wrong?
- Answer:
- When price of a good ↓, the substitution effect is positive, the income effect is often either positive or negative but relatively small in magnitude, so the total effect is still positive.
- Only when the income effect is negative and big enough in magnitude, the law of demand can be violated.

Bonus Quizzes

Question 1 0.5 pts

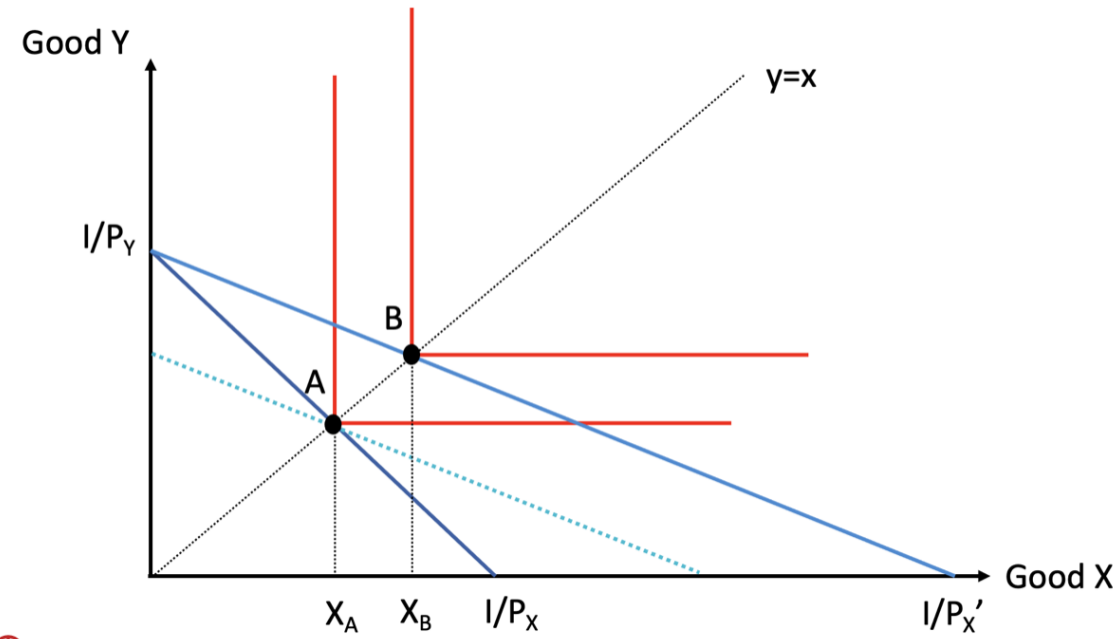
A change in consumption of a good resulting from a change in its relative price, with utility level held constant, is referred to as _____.

- the substitution effect
- the income effect
- the total effect of a price change
- the wealth effect

Question 2

0.5 pts

Suppose Good X and Y are perfect complements. The kinks of indifference curves fall on the 45-degree line. When the price of Good X decreases from P_X to P_X', the total effect of the price change on consumption of Good X is the same with _____.



- the substitution effect
- the income effect
- the wealth effect